



Login Management System (Industry Style)

JSP + Servlet + JDBC + MySQL + Maven + Bootstrap 5 + Session Management

Project Overview

Features

- ✓ Attractive Login UI
 - ✓ Bootstrap 5
 - ✓ Font Awesome Icons
 - ✓ Show/Hide Password
 - ✓ Remember Me Checkbox
 - ✓ MySQL Database Validation
 - ✓ Session Management
 - ✓ Dashboard Page
 - ✓ Profile Card
 - ✓ Logout Functionality
 - ✓ Session Timeout
-

STEP 1 : Create Maven Project

File



New



Maven Project



Create Simple Project



Next

Enter

Group Id : com.login
Artifact Id : LoginSystem
Version : 1.0
Packaging : war

Finish.

STEP 2 : Project Structure

Create Packages

src/main/java

com.login.db

com.login.model

com.login.dao

com.login.controller

Create JSP Pages

src/main/webapp

login.jsp

dashboard.jsp

Final Structure

LoginSystem

src/main/java

```
|— com.login.db
|   DBConnection.java
|
|— com.login.model
|   User.java
|
|— com.login.dao
```

```
|   UserDao.java
|
└── com.login.controller
    LoginServlet.java
    LogoutServlet.java
```

src/main/webapp

```
├── login.jsp
├── dashboard.jsp
└── WEB-INF
```

STEP 3 : Add Dependencies

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0">
```

```
    <modelVersion>4.0.0</modelVersion>
```

```
    <groupId>com.login</groupId>
```

```
    <artifactId>LoginSystem</artifactId>
```

```
    <version>1.0</version>
```

```
    <packaging>war</packaging>
```

```
    <dependencies>
```

```
        <dependency>
```

```
            <groupId>mysql</groupId>
```

```
            <artifactId>mysql-connector-j</artifactId>
```

```
            <version>9.3.0</version>
```

```
        </dependency>
```

```
        <dependency>
```

```
            <groupId>jakarta.servlet</groupId>
```

```
            <artifactId>jakarta.servlet-api</artifactId>
```

```
            <version>6.1.0</version>
```

```
            <scope>provided</scope>
```

```
        </dependency>
```

```
    </dependencies>
```

</project>

Maven Update

Right Click Project



Maven



Update Project

STEP 4 : Create Database

```
create database logindb;
```

```
use logindb;
```

```
create table users
```

```
(
```

```
id int primary key auto_increment,
```

```
username varchar(50),
```

```
password varchar(50)
```

```
);
```

```
insert into users(username,password)
```

```
values('admin','1234');
```

STEP 5 : Create DBConnection

Package

```
com.login.db
```

DBConnection.java

```
package com.login.db;
```

```
import java.sql.Connection;
```

```
import java.sql.DriverManager;
```

```
public class DBConnection {
```

```
public static Connection getConnection() {  
  
    Connection con = null;  
  
    try {  
  
        Class.forName("com.mysql.cj.jdbc.Driver");  
  
        con = DriverManager.getConnection(  
            "jdbc:mysql://localhost:3306/logindb",  
            "root",  
            "root");  
  
    } catch (Exception e) {  
        e.printStackTrace();  
    }  
  
    return con;  
}  
}
```

STEP 6 : Create Model Class

Package

com.login.model

User.java

```
package com.login.model;
```

```
public class User {
```

```
    private String username;  
    private String password;
```

```
    public String getUsername() {  
        return username;  
    }
```

```
    public void setUsername(String username) {  
        this.username = username;  
    }
```

```
public String getPassword() {  
    return password;  
}  
  
public void setPassword(String password) {  
    this.password = password;  
}  
}
```

STEP 7 : Create DAO

Package

com.login.dao

UserDAO.java

```
package com.login.dao;
```

```
import java.sql.*;
```

```
import com.login.db.DBConnection;
```

```
public class UserDAO {
```

```
    public boolean validateUser(String username,  
                                String password) {
```

```
        boolean status = false;
```

```
        try {
```

```
            Connection con =  
                DBConnection.getConnection();
```

```
            String sql =  
                "select * from users where username=? and password=?";
```

```
            PreparedStatement ps =  
                con.prepareStatement(sql);
```

```
            ps.setString(1, username);  
            ps.setString(2, password);
```

```

        ResultSet rs =
            ps.executeQuery();

        if(rs.next()) {
            status = true;
        }

    } catch(Exception e) {
        e.printStackTrace();
    }

    return status;
}
}

```

STEP 8 : Create Login Servlet

Package

com.login.controller

LoginServlet.java

```
package com.login.controller;
```

```
import java.io.IOException;
```

```
import com.login.dao.UserDAO;
```

```
import jakarta.servlet.annotation.WebServlet;
```

```
import jakarta.servlet.http.*;
```

```
@WebServlet("/login")
```

```
public class LoginServlet extends HttpServlet {
```

```
    protected void doPost(
        HttpServletRequest request,
        HttpServletResponse response)
        throws IOException {
```

```
        String username =
            request.getParameter("username");
```

```

String password =
    request.getParameter("password");

UserDAO dao =
    new UserDAO();

if(dao.validateUser(username,password)) {

    HttpSession session =
        request.getSession();

    session.setAttribute("user",username);

    session.setMaxInactiveInterval(300);

    response.sendRedirect("dashboard.jsp");

} else {

    response.sendRedirect(
        "login.jsp?error=1");
    }
}
}

```

STEP 9 : Create Logout Servlet

LogoutServlet.java

```

package com.login.controller;

import java.io.IOException;

import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.*;

@WebServlet("/logout")
public class LogoutServlet
    extends HttpServlet {

    protected void doGet(
        HttpServletRequest request,
        HttpServletResponse response)
        throws IOException {

```

```
        HttpSession session =  
            request.getSession();  
  
        session.invalidate();  
  
        response.sendRedirect(  
            "login.jsp");  
    }  
}
```

STEP 10 : Create Attractive login.jsp

```
<%@ page language="java" %>  
<!DOCTYPE html>  
<html>  
<head>  
  
<title>Login System</title>  
  
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"  
rel="stylesheet">  
  
<link rel="stylesheet"  
href="https://cdn.jsdelivr.net/npm/font-awesome@6.5.1/css/all.min.css">  
  
<style>  
  
body{  
    height:100vh;  
    display:flex;  
    justify-content:center;  
    align-items:center;  
    background:linear-gradient(135deg,#667eea,#764ba2);  
}  
  
.login-card{  
    width:420px;  
    padding:40px;  
    border-radius:20px;  
    background:rgba(255,255,255,.15);  
    backdrop-filter:blur(15px);  
    box-shadow:0 8px 32px rgba(0,0,0,.3);  
    color:white;
```

```
}

</style>

</head>

<body>

<div class="login-card">

<h2 class="text-center mb-4">

<i class="fa-solid fa-user-lock"></i>

Login

</h2>

<%
if(request.getParameter("error")!=null){
%>

<div class="alert alert-danger">
Invalid Username Or Password
</div>

<%
}
%>

<form action="login" method="post">

<div class="mb-3">

<label>Username</label>

<input type="text"
name="username"
class="form-control">

</div>

<div class="mb-3">

<label>Password</label>

<div class="input-group">
```

```
<input type="password"
id="pass"
name="password"
class="form-control">
```

```
<button type="button"
class="btn btn-light"
onclick="showPass()">
```



```
</button>
```

```
</div>
```

```
</div>
```

```
<div class="form-check mb-3">
```

```
<input type="checkbox"
class="form-check-input">
```

```
<label>
Remember Me
</label>
```

```
</div>
```

```
<button class="btn btn-success w-100">
```

Login

```
</button>
```

```
</form>
```

```
</div>
```

```
<script>
```

```
function showPass(){

var x=document.getElementById("pass");

if(x.type==="password")
{
x.type="text";
}
}
```

```
else
{
x.type="password";
}

}

</script>

</body>
</html>
```

STEP 11 : Create Attractive dashboard.jsp

```
<%@ page language="java" %>

<%
String user =
(String)session.getAttribute("user");

if(user==null)
{
response.sendRedirect("login.jsp");
return;
}
%>

<!DOCTYPE html>
<html>
<head>

<title>Dashboard</title>

<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">

<style>

body{
background:#f5f6fa;
}
```

```
.card-box{  
margin-top:100px;  
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<nav class="navbar navbar-dark bg-primary">
```

```
<div class="container-fluid">
```

```
<span class="navbar-brand">
```

Login System

```
</span>
```

```
<a href="logout"  
class="btn btn-danger">
```

Logout

```
</a>
```

```
</div>
```

```
</nav>
```

```
<div class="container card-box">
```

```
<div class="card shadow p-4">
```

```
<h2>
```

Welcome

```
<%=user%>
```



```
</h2>
```

```
<hr>
```

```
<div class="row">
```

```
<div class="col-md-4">

<div class="card bg-success text-white p-3">

Users

</div>

</div>

<div class="col-md-4">

<div class="card bg-warning text-white p-3">

Reports

</div>

</div>

<div class="col-md-4">

<div class="card bg-info text-white p-3">

Settings

</div>

</div>

</div>

</div>

</div>

</div>

</div>

</body>
</html>
```

Final Flow

User Opens Login Page



Enters Username Password



LoginServlet



UserDAO



MySQL Database



Valid User Found



Session Created



dashboard.jsp



Logout Button



LogoutServlet



Session Destroyed



login.jsp

Test Credentials

Username : admin

Password : 1234